English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Mathematics / Applied Mathematics, 8 June 2024 Test 2 ___*The exact grading formula will depend on the results. As an approximation, use 2 + the number of points. Time allowed: 4 hours. Good luck!*

Problem 1. Based on previous games, Angel models his score as a random variable with mean 5011 points and variance 4000. Approximately how many games does he need to be at least 99% certain that:

1. (0.5 points) his total score is at least one million?
2. (0.5 points) his highest score among these games is at least 10000?

Problem 2. (1 point) Let ξ, η be independent, with $\xi \sim \text{Exp}(2)$ and $\eta \sim U(0,3)$, that is,

$$f_\xi(x) = \begin{cases} 2e^{-2x}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad f_\eta(x) = \begin{cases} 1/3, & 0 < x < 3, \\ 0, & \text{otherwise}. \end{cases}$$

Find their correlation, $\mathbb{P}(\xi < \eta)$ and the density of ξ/η.

Problem 3.
1. (0.25 points) Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, and suppose a sample of n observations is available. Test

$$H_0 : \mu = 3.8 \qquad \text{against} \qquad H_1 : \mu \neq 3.8$$

at significance level $\alpha = 0.05$. Can H_0 be rejected if $n = 16$, $\overline{X} = 3.295$ and $S^2 = 1.4641$? Derive the test using likelihood ratios.

2. (0.5 points) Let $X \sim U(\alpha, \beta)$. Find method of moments and maximum likelihood estimators of α, β. Determine whether they are unbiased and/or consistent.

Problem 4. Let X, Y be independent Exp(1) variables and $Z = \sqrt{X/Y}$.

1. (0.5 points) Find F_Z and use it to calculate $\mathbb{E}Z$.
2. (0.25 points) Express $\mathbb{E}Z$ using the joint density of X, Y. Conclude that $\Gamma(1/2) = \sqrt{\pi}$.[1]

Problem 5. The aim is to find the integer moments of $X \sim N(\mu, \sigma^2)$. For convenience, begin with $Z \sim N(0,1)$.

1. (0.25 points) Find $\mathbb{E}[e^{\lambda Z}]$ for $\lambda > 0$ and $\mathbb{E}[Z^k]$ for $k \leq 6$.
2. (0.25 points) Prove $\mathbb{E}[f(Z)Z] = \mathbb{E}[f'(Z)]$ for functions for which both expectations are well defined.
3. (0.25 points) Find $\mathbb{E}[Z^k]$ for every $k \geq 0$. Can you obtain the corresponding result for $\mathbb{E}[X^k]$?
4. (0.25 points) Find $\mathbb{P}(Z^k > Z^{k-1})$ for $k \geq 5$.

Problem 6. A vector $X = (X_1, \ldots, X_n)$ is normally distributed, or Gaussian, with mean $\mu = (\mu_1, \ldots, \mu_n)$ and covariance matrix Σ, $\Sigma_{ij} = \text{Cov}(X_i, X_j)$, if every linear combination $\langle \lambda, X \rangle = \sum_i \lambda_i X_i$ has distribution $N(\langle \lambda, \mu \rangle, \sum_{i,j} \lambda_i \lambda_j \text{Cov}(X_i, X_j))$.[2]

1. (0.10 points) Find the moment generating function $M_Y(t)$ of $Y \sim N(\mu, \sigma^2)$.
2. (0.25 points) Using the assertion that $M_Y(t) M_W(t) = M_{Y+W}(t)$ in a nonempty neighbourhood of 0 implies independence of Y, W, and its analogue for more than two variables, prove that uncorrelated components of a Gaussian vector are jointly independent normally distributed variables.

   For the rest of the problem, assume $X = (X_1, \ldots, X_n)$ consists of independent $N(0, \sigma^2)$ variables.

3. (0.25 points) Let P be an $n \times n$ orthogonal matrix, so $PP^T = I_n$. Prove that PX also consists of independent $N(0, \sigma^2)$ variables.
4. (0.25 points) Let V be a k-dimensional linear subspace of $\mathbb{R}^n$, and $V^\perp$ its orthogonal complement, so $\mathbb{R}^n = V \oplus V^\perp$. Prove that the projections of X onto V and $V^\perp$ are independent Gaussian vectors and that their lengths are independent and chi-squared distributed with parameters k and $n-k$, respectively.

   Using the preceding part, explain why it suffices to consider $X_V = (X_1, \ldots, X_k, 0, \ldots, 0)$ and $X - X_V$.

[1] Euler's gamma function is $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} \, \mathrm{d}t$.

[2] The sum is the quadratic form $\lambda \Sigma \lambda^T$.

5. (0.25 points) Take V to be the subspace of $\mathbb{R}^n$ spanned by $(1, \ldots, 1)$. Prove that

$$\overline{X} = \frac{X_1 + \cdots + X_n}{n} \qquad \text{and} \qquad (n-1)S^2 = \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$$

are independent, and specify their distributions.

Problem 7. *Bonus* (0.5 points) Find $\lim_{n\to\infty} e^{-n} \left(\frac{n}{1!} + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!} \right)$.